Problem Statement

For a positive integer n, define

$$f(n) = n + \frac{16 + 5n - 3n^2}{4n + 3n^2} + \frac{32 + n - 3n^2}{8n + 3n^2} + \cdots + \frac{25n - 7n^2}{7n^2}$$

Evaluate the following limit:

$$\lim_{n\to\infty} f(n)$$

Detailed Solution

This limit can be evaluated by identifying the general term of the summation, simplifying the expression to cancel out the leading n term, and then transforming the resulting sum into a Riemann sum that converges to a definite integral. Let's break this down step-by-step.

Step 1: Identify the general term of the summation

Let's analyze the terms in the summation of $f(n)$. We can write:

$$f(n) = n + \sum_{k=1}^{N} T_k$$

where T_k represents the k-th term in the sequence of fractions:

- $T_1 = \frac{16+5n-3n^2}{4n+3n^2}$
- $T_2 = \frac{32+n-3n^2}{8n+3n^2}$
- $T_{\text{last}} = \frac{25n-7n^2}{7n^2}$

Looking at the denominators, we see an arithmetic progression of the form $4kn + 3n^2$. Setting this general expression equal to the last denominator, we have:

$$4kn + 3n^2 = 7n^2 \implies 4kn = 4n^2 \implies k = n$$

Thus, there are exactly n terms in the summation, meaning the sum runs from $k = 1$ to n.

Next, let's determine the general formula for the numerator, N_k, as a function of k and n. Since the constant term is $16k$ and the coefficient of n^2 is always -3, we can propose the form:

$$N_k = 16k + a_k n - 3n^2$$

For $k = 1$, we have $a_1 = 5$. For $k = 2$, we have $a_2 = 1$. Since a_k decreases linearly by 4 for each step, we can write:

$$a_k = 5 - 4(k - 1) = 9 - 4k$$

To verify this formula for the final term when $k = n$:

$$N_n = 16n + (9 - 4n)n - 3n^2 = 16n + 9n - 4n^2 - 3n^2 = 25n - 7n^2$$

This perfectly matches the final term's numerator. Therefore, the general term of our summation is:

$$T_k = \frac{16k + (9 - 4k)n - 3n^2}{4kn + 3n^2}$$

Step 2: Decompose and simplify the general term

We can algebraicly manipulate the numerator of T_k to group terms sharing a common factor of n:

$$T_k = \frac{(16k - 4kn) + 9n - 3n^2}{4kn + 3n^2} = \frac{-n(4k + 3n) + 16k + 9n}{n(4k + 3n)}$$

Splitting this fraction yields:

$$T_k = \frac{-n(4k + 3n)}{n(4k + 3n)} + \frac{16k + 9n}{n(4k + 3n)} = -1 + \frac{16k + 9n}{n(4k + 3n)}$$

Now, substituting T_k back into the expression for $f(n)$, we obtain:

$$\begin{aligned} f(n) &= n + \sum_{k=1}^{n} \left[-1 + \frac{16k + 9n}{n(4k + 3n)}\right] \\ &= n - \left(\sum_{k=1}^{n} 1\right) + \sum_{k=1}^{n} \frac{16k + 9n}{n(4k + 3n)} \\ &= n - n + \sum_{k=1}^{n} \frac{16k + 9n}{n(4k + 3n)} \\ &= \sum_{k=1}^{n} \frac{16k + 9n}{n(4k + 3n)} \end{aligned}$$

The leading n term is eliminated, leaving us with a clean summation.

Step 3: Formulate the Riemann sum

To express this limit as a definite integral, we rewrite the general term in terms of the variable $x_k = \frac{k}{n}$ with width $\Delta x = \frac{1}{n}$. We divide both the numerator and the denominator of the summand by n:

$$\frac{16k + 9n}{n(4k + 3n)} = \frac{16\left(\frac{k}{n}\right) + 9}{n\left(4\left(\frac{k}{n}\right) + 3\right)} = \frac{1}{n} \cdot \frac{16\left(\frac{k}{n}\right) + 9}{4\left(\frac{k}{n}\right) + 3}$$

Thus, we can represent $f(n)$ as:

$$f(n) = \sum_{k=1}^{n} \frac{1}{n} g\left(\frac{k}{n}\right)$$

where our function $g(x)$ is defined as:

$$g(x) = \frac{16x + 9}{4x + 3}$$

Step 4: Convert to a definite integral and evaluate

As $n \to \infty$, the Riemann sum converges to the definite integral of $g(x)$ over the interval $[0, 1]$:

$$L = \lim_{n\to\infty} f(n) = \int_0^1 \frac{16x+9}{4x+3}\, dx$$

To compute this integral, we rewrite the numerator to separate the constant term:

$$\frac{16x+9}{4x+3} = \frac{4(4x+3)-3}{4x+3} = 4 - \frac{3}{4x+3}$$

Now, integrating term-by-term:

$$\begin{aligned} L &= \int_0^1 \left(4 - \frac{3}{4x+3}\right) dx \\ &= \left[4x - \frac{3}{4}\ln|4x+3|\right]_0^1 \end{aligned}$$

Evaluating this at the integration boundaries:

$$\begin{aligned} L &= \left(4(1) - \frac{3}{4}\ln|4(1)+3|\right) - \left(4(0) - \frac{3}{4}\ln|4(0)+3|\right) \\ &= \left(4 - \frac{3}{4}\ln(7)\right) - \left(0 - \frac{3}{4}\ln(3)\right) \\ &= 4 - \frac{3}{4}(\ln(7) - \ln(3)) \\ &= 4 - \frac{3}{4}\ln\left(\frac{7}{3}\right) \end{aligned}$$

Final Answer:

$$\mathbf{4 - \frac{3}{4}\ln\left(\frac{7}{3}\right)}$$